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# Berdakh Abibullaev, Ph.D.

*Curriculum Vitae*

## Research Area

**Neural Engineering:** Brain-Computer/Machine Interfaces; EEG/MEG/ECoG/fNIRS signal processing.

**Machine Learning:** Neural Networks and Deep Learning, Kernel based learning, Support vector machines.

## Education

02/2010 **Ph.D**, *Yeungnam University*, Department of Electronic Engineering, South Korea.
- *Research area* : EEG time-series prediction, Machine learning and Neural Networks;

08/2006 **M.Sc**, *Yeungnam University*, Department of Electronic Engineering, South Korea.
- *Research area*: Wavelet Analysis, Time-frequency localization; Physiological feature extraction;

07/2004 **B.Sc**, *Tashkent University of Information Technologies*, Uzbekistan.

## Academic Work Experience

09/2015-present **Assistant Professor**, *Robotics & Mechatronics Department, School of Engineering and Digital Science, Nazarbayev University*, Nur-Sultan, Kazakhstan.

Research Focus:
- Deep learning algorithms for Brain-Computer/Machine Interfaces.
- Brain-Computer/Machine Interfaces in Neural-Rehabilitation after Stroke.
- Transfer learning applications in session/subject independent Brain-Computer/Machine Interfaces.
- High-Level Robotic Manipulator Control via Brain-Computer Interfaces for Assistive Applications

5/2018 - 7/2018 **Visiting Professor**, *Department of Electrical Engineering & Computer Science, University of Houston*, Houston, Texas, USA.
- Cortical source imaging of electroencephalographic signals in patients with stroke to examine neural correlates brain plasticity pre and post clinical trials with BMI. actuated exoskeleton.

9/2014 - 9/2015 **NIH Postdoctoral Research Fellow II**, *Department of Electrical Engineering & Computer Science, University of Houston*, Houston, Texas, USA.
- BMI-Exo Project: Brain-Machine Interface Control of a Therapeutic Exoskeleton for Neurorehabilitation.
  - Estimating neural correlates and mechanisms of an external command (mental intent) used to drive the external therapeutic exoskeleton for BMI.
- Neural Correlates of Infant Mirror Neuron System
  - Investigate the role the mirror neuron system plays in the motor learning of young infants via noninvasively acquired brain signals;
  - Categorizations of large-scale brain connectivity analysis to understand neural mechanisms of infant behavior.
- Electromagnetic-source localization and estimating brain network in connectivity analysis

01/2014-09/2014 **Research Scientist**, *Department of Neurology, Samsung Medical Center*, Seoul, South Korea.
- Investigated High-Frequency Oscillations of ECoG/EEG signals relevant to Epileptogenesis using machine learning algorithms. Supported medical doctors through brain data analysis in the presurgical evaluation of patients with refractory epilepsy.
- Neuroelectromagnetic source localization from scalp EEG & MEG recordings for delineation of epileptogenic zones in the surgical treatment of refractory focal epilepsy.

04/2011-12/2013 **Senior Research Scientist**, *Robotics Research Division, Daegu Gyeongbuk Institute of Science and Technology*, South Korea.

Development of rehabilitation strategies for stroke patients via robot-assisted physical therapy with BCIs.
- Implemented kernel-based machine learning algorithms for online BCIs to control neuro-rehabilitation devices.
- Examined inter-subject/session variabilities to enhance fNIRS-based BCI performances.
- Design and optimization of non-invasive based BCI paradigms for real-time robot control.

01/2010-04/2011 **Postdoctoral Research Scholar**, *Robotics Research Division, Daegu Gyeongbuk Institute of Science and Technology*, South Korea.

Brain-Computer Interfaces in therapy and diagnosis of Attention Deficit and Hyperactivity Disorders (ADHD).
- Involved in quantitative EEG and Neurofeedback methods of ADHD children. Time-frequency localization; Identification of strongly relevant neural signal patterns of ADHD using machine learning.
- Developed machine learning-based semi-supervised feature selection and classification algorithms.
- Participated in the developmental assessment of ADHD children during a long-term neuro-therapy in cooperation with a neurorehabilitation center at Daegu University.

## Honors and Awards

11/2017 Best paper award: *fNIRS motion artifact correction for overground walking using entropy-based unbalanced optode decision and wavelet regression neural network*," in Multisensor Fusion and Integration for Intelligent Systems, 2017 IEEE International Conference.

06/2014 Best poster award on "*The evaluation of accuracy and clinical usefulness of 3D EEG source localization analysis*," Korean Epilepsy Congress 2014.

12/2012 Best Researcher of the Year Award. Robotics Research Division, Daegu Gyeongbuk Institute of Science and Technology, Daegu, South Korea.

01/2011 Enlisted in Marquis Who's Who in the World 2011, 28th Edition, USA

09/2006 Korean Government Information Technology Scholarship for International Graduate Students for the Ph.D. program, South Korea.

09/2004 Brain Korea #21 Scholarship for the MSc program.

## Technical Skills

**Software skills**.
Expert in the Python language and libraries such as Cython; very good knowledge of C, C++. Experience with various tools and languages, including bash, LATEX, HTML, Git, and different database query languages.

**Data Science**.
Expert knowledge in NumPy, Scipy, Pandas, Sklearn, PyTorch Deep larning libraries

**Neural data acquisition systems**.
Expertise with Nicolet EEG Amplifiers, Blackrock Microsystems (intracranial recording), Functional Near-Infrared Spectroscopy FOIRE3000, BIOPAC fNIR200 Optical Brain Imaging, BioSemi Active-Two EEG system, Brain Products EEG amp., BIOPAC MP100 EEG amp. And G.tec g.USBamp systems.

**Productivity applications**.
TEX: [LATEX, BibTEX, PSTricks, Beamer].

**System Environments**.
Linux (Pop_OS, Ubuntu), Windows

## Service to the Scientific Community

**Associate Editor**.
- IEEE Access Journal (Impact Factor: 3.7)
- PeerJ Computer Science (Impact Factor: 3.09)

**Occasional Reviewer**.
- IEEE Access Journal
- Neuroimage, Elsevier.
- IEEE Transactions on Neural Networks and Learning Systems
- Brain Research, Elsevier.

**Program Committee Member and Session Chair**.
- International Workshop on Artificial Intelligence for Healthcare Applications at International Conference on Pattern Recognition 2020.
- Session Chair: The 6th International Conference on Brain-Computer Interface (BCI), IEEE, 2018, South Korea.
- Session Chair: IEEE International Conference on Multisensory Fusion and Integration for Intelligent Systems, 2017, Daegu, South Korea.
- Session Chair: International Conference on Control Automation and Systems Oct.27-30, 2010, Seoul, South Korea

## Project Proposals, Participation & Coordination

01/2021 **Principal Investigator**, ***Brain-Machine Interfaces for Neural Rehabilitation after Stroke***, This project aims to accelerate the efficacy of robotic rehabilitation using a new neural interface system developed for upper-limb repair after stroke. It aims to facilitate expediate motor recovery from incomplete upper-limb paralysis and investigate in-depth understanding of cortical reorganization in the brain associated with the recovery..

01/2021 **Co-Investigator**, ***Classifier Validity and Its Applications in Bioinformatics***, The project studies the measures of classifier validity and explores their applications in bioinformatics. Nazarbayev University..

01/2019 **Co-Investigator**, ***Neural-assisted optimization of human learning with multimodal neuroimaging of EEG and NIRS***, Implementing and validate a neural-assisted, machine-learning-based intelligent tutoring system that uses information from physiological signals to adapt the learning schedule to a user's individual learning capacity..

09/2015 **Principal Investigator**, ***Development of a Brain-Computer Interface with a Robot and Functional-Electrical Stimulation for Upper-Limb Stroke Rehabilitation (BCI-ROBOFES)***, Evaluated by the committee of Oak Ridge Associated Universities (ORAU);.

9/2014 **Participant**, *Cognitive-Motor Development Research in Infants.*, Lead a team of Ph.D. students in exploring & decoding of the neural basis of *Imitation and Action Understanding* in freely-behaving infants via multimodal brain imaging techniques, University of Houston, Texas.

9/2014 **Participant**, ***The BMI-Exo Project: Facilitating Personalized Robotic Rehabilitation Of The Upper Limb, [NIH Grant R01NS081854-02, National Robotics Initiative]***, Lead a part of research that focuses on understanding neurophysiological correlates of motor learning during movements in stroke patients..

04/2014 **Participant**, ***Development of high-resolution subdural electrodes and data mining methods for precise localization of epileptogenic zone in treatment of patients with refractory epilepsy.***, Participated in idea generation and in drafting/writing of the project proposal at Samsung Medical Center, Korea.

# Publications

## Patents

[1] J. An, S.H. Jin, S.H. Lee, J.I. Moon B. Abibullaev, J.H. Ahn and G.H. Jang. **Rehabilitation Training System and Method**. *U.S. Patent No*: 9,081,890,Washington, DC: United States. Patent and Trademark Office, 2015 (https://www.google.com/patents/US9081890).

[2] J. An, S.H. Jin, S.H. Lee, J.I. Moon B. Abibullaev, J.H. Ahn and G.H. Jang. **Self-directed rehabilitation training method combining brain signals and functional electro-stimulation**. *EFS ID*: 17077057, 09-OCT-2013, United States, (http://www.google.com/patents/US20140200632).

[3] J. An, S.H. Jin, S.H. Lee, B. Abibullaev, J.I. Moon , J.H. Ahn and G.H. Jang. **Combining Brain Signals and Functional Electrostimulation Self-Directed Rehabilitation Method**, Patent Registration Number: 101501524000, 05.03.2015. South Korea,(http://eng.kipris.or.kr/).

## Refereed Journal Publications

[1] B. Abibullaev and A. Zollanvari, **A Systematic Deep Learning Model Selection for P300-Based Brain-Computer Interfaces**, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, DOI: 10.1109/TSMC.2021.3051136, 2021.(https://ieeexplore.ieee.org/document/9351887)*(**Impact Factor: 13.451**; Quartile rank: Q1).*

[2] A. Zollanvari and B. Abibullaev **Bias Correction for Linear Discriminant Analysis**, *Pattern Recognition Letters*, vol. 151, pp. 41-47, 2021, https://doi.org/10.1016/j.patrec.2021.07.026 *(Impact Factor: 3.7; Quartile rank: Q1).*

[3] B. Abibullaev and A. Zollanvari, **Learning Discriminative Spatiospectral Features of ERPs for Accurate Brain-Computer Interfaces** *IEEE Journal of Biomedical and Health Informatics*, vol. 98, pp. 1-12, 2019, DOI: 10.1109/JBHI.2018.2883458 *(Impact Factor: 5.772; Quartile rank: Q1).*

[4] B. Abibullaev, I. Dolzhikova and A. Zollanvari, **A Brute-force CNN Model Selection for Accurate Classification of Sensorimotor Rhythms in BCIs** *IEEE Access*, 2020, DOI: 10.1109/ACCESS.2020.2997681 *(Impact Factor: 3.36; Quartile rank: Q1).*

[5] B. Abibullaev, A. Zollanvari, B. Saduanov and T. Alizadeh, **Design and Optimization of a BCI-Driven Telepresence Robot Through Programming by Demonstration** *IEEE Access*, August 2019, DOI: 10.1109/ACCESS.2019.2933268 *(Impact Factor: 3.36; Quartile rank: Q1).*

[6] A. Zollanvari, M. Abdirash, A. Dadlani and B Abibullaev. **Asymptotically Bias-Corrected Regularized Linear Discriminant Analysis for Cost-Sensitive Binary Classification** *IEEE Signal Processing Letters*, 22 May 2019, DOI: 10.1109/LSP.2019.2918485 *(Impact Factor: 3.1; Quartile rank: Q1).*

[7] B. Abibullaev, J An, SH Lee, JI Moon. **Design and Evaluation of Action Observation and Motor Imagery based BCIs using NIRS** *Measurement*,vol. 98, pp. 250-261, 2017, Elsevier *(Impact Factor: 3.927; Quartile rank: Q1).*

[8] N.A. Bhagat, A. Venkatakrishnan, B. Abibullaev, E.J. Artz, N. Yozbatiran, A. Blank, J. French, C. Karmonik, R.G.Grossman, M.K O'Malley, G. Francisco, J.L. Contreras-Vidal. **Design and optimization of an EEG-based brain-machine interface (BMI) to an upper-limb exoskeleton for stroke survivors.** *Frontiers in Neuroscience*, vol. 10, March, 2016 (Impact Factor: 4.677; Quartile rank: Q1).

[9] J.G. Cruz-Garza, Z.R. Hernandez, T. Tse, E. Caducoy, B. Abibullaev, J.L. Contreras-Vidal. **A novel experimental and analytical approach to the multimodal neural decoding of intent during social interaction in freely-behaving human infants.** *Journal of Visualized Experiments*, DOI: 10.3791/53406, October, 2015 (Impact Factor : 1.5; Quartile rank: Q2).

[10] C.H. Park, J.H Seo, D. Kim, B. Abibullaev, H. Kwon, Y.H. Lee, M.Y. Kim, K. Kim, J.S. Kim, E.Y. Joo, S.B. Hong, (2015, Feb). **EEG Source Imaging in Partial Epilepsy in Comparison with Presurgical Evaluation and**

**Magnetoencephalography.** *Journal of Clinical Neurology*, 2015 Feb 17, 11:e12 (Impact Factor : 3.077; Quartile rank: Q2).

[11] B. Abibullaev, J An, S.H. Jin, and J.I. Moon. **Classification of brain hemodynamic signals arising from visual action observation tasks for brain-computer interfaces : An fNIRS study.** *Measurement*, DOI:10.1016/j.measurement.2013.11.044, 2014. Elsevier (Impact Factor : 3.927; Quartile rank: Q1).

[12] B. Abibullaev, J An, S.H. Lee, S.H. Jin, and J.I. Moon. **Minimizing inter-subject variability in fNIRS based brain computer interfaces via multiple-kernel support vector learning.** DOI: 10.1016/j.medengphy.2013.08.009, *Medical Engineering Physics*, 2013. Elsevier (Impact Factor : 2.242; Quartile rank: Q2).

[13] B. Abibullaev and J. An. **Classification of frontal cortex hemodynamic response during cognitive tasks using wavelet transforms and machine learning algorithms.** *Medical Engineering Physics*, 34(10):1394–410, 2012. Elsevier (Impact Factor : 2.242; Quartile rank: Q2).

[14] B. Abibullaev and J. An. **Decision support algorithm for diagnosis of ADHD disorder using electroencephalograms.** *Journal of Medical Systems*, 36(4):2675–2688, 2011. Springer (Impact Factor : 4.460; Quartile rank: Q2).

[15] B. Abibullaev, J. An, and J.I. Moon. **Neural network classification of brain hemodynamic responses from four mental tasks.** *Int J Optomechatronics*, 5(4):340–359, 2011. Taylor & Francis (Impact Factor : 1.9; Quartile rank: Q2).

[16] B. Abibullaev and H.D. Seo. **A new QRS detection method using wavelets and artificial neural networks.** *Journal of Medical Systems*, 35(4):683–691, 2011. Springer (Impact Factor : 4.460; Quartile rank: Q2).

[17] B. Abibullaev, M.S. Kim, and H.D. Seo. **Epileptic spike detection using continuous wavelet transforms and artificial neural networks.** *Int J of Wavelets, Multires and Inform Proc*, 8(1):33–48, 2010. Worldscientific (Impact Factor : 0.463; Quartile rank: Q3).

[18] B. Abibullaev, M.S. Kim, and H.D. Seo. **Seizure detection in temporal lobe epileptic EEGs using the best basis wavelet functions.** *Journal of Medical Systems*, 34(4):755–765, 2010. Springer (Impact Factor : 4.460; Quartile rank: Q2).

[19] M.S. Kim, Y.C. Cho, B. Abibullaev, and H.D. Seo. **Analysis of brain functions and classification of sleep stage EEG using daubechies wavelet.** *Sensors and Materials*, 20(1):1–15, 2008. MYU, Japan (Impact Factor: 0.92; Quartile rank: Q3).

## Conference proceedings

[1] I. Dolzhikova, B. Abibullaev, A. Zollanvari, "**An Ensemble of Convolutional Neural Networks for Zero-Calibration ERP-Based BCIs**", 2022 10th International Winter Conference on Brain-Computer Interface (BCI), 2022, pp. 1-4, doi: 10.1109/BCI53720.2022.9734904.

[2] I. Dolzhikova, B. Abibullaev, R. Sameni, A. Zollanvari, "**An Ensemble CNN for Subject-Independent Classification of Motor Imagery-based EEG**", the 43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBS), October 31 – November 4, 2021

[3] Y. Kainolda, B. Abibullaev, R. Sameni, A. Zollanvari, "**Is Riemannian Geometry Better than Euclidean in Averaging Covariance Matrices for CSP-based Subject-Independent Classification of Motor Imagery?**", the 43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBS), October 31 – November 4, 2021

[4] M. Nurpeiissov, B. Abibullaev, T. Alizadeh, "**A novel human-robot interaction framework based on Telegram and programming by demonstration**", 9th International Conference on Robot Intelligence Technology and Applications (RITA 2021), on Dec 16 - 17, 2021 at KAIST Daejeon, Korea.

[5] S. Abilkassov, M. Kairgaliyev, B. Zhakanov, B. Abibullaev, "**A System For Drivers' Cognitive Load Estimation Based On Deep Convolutional Neural Networks and Facial Feature Analysis**", in 22nd IEEE International Conference on Industrial Technology, Virtual 10-12, 2021, Spain.

[6] K. Kunanbayev, B. Abibullaev, A. Zollanvari, "**Data Augmentation for P300-based Brain-Computer Interfaces Using Generative Adversarial Networks**", in 2021 9th International Winter Conference on Brain-Computer Interface (BCI), 22-24 Feb. 2021, pp. 1-7, Gangwon, Korea (South), doi: 10.1109/BCI51272.2021.9385317.

[7] K. Kunanbayev, D. Azhigulov, B. Abibullaev, A. Zollanvari, "**Deep Transfer Learning for Subject-Independent ERP-based BCIs**", in 2021 9th International Winter Conference on Brain-Computer Interface (BCI), 22-24 Feb. 2021, pp. 1-3, Gangwon, Korea (South), doi: 10.1109/BCI51272.2021.9385311.

[8] A. Oleinikov, B. Abibullaev, M. Folgheraiter, "**On the Classification of Electromyography Signals to Control a Four Degree-Of-Freedom Prosthetic Device**," in 2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), July 20-24, Montreal, Canada.

[9] B. Saduanov, D. Tokmurzina, K. Kunanbayev and B. Abibullaev, "**Design and Optimization of a Real-Time Asynchronous BCI Control Strategy for Robotic Manipulator Assistance**", in 2020 8th International Winter Conference on Brain-Computer Interface. (BCI), 2020, GangWon, South Korea.

[10] Adilet Tuleuov and B. Abibullaev, "**Deep Learning Models for Subject-Independent ERP-based Brain-Computer Interfaces**," in *the 2019 9th International IEEE/EMBS Conference on Neural Engineering (NER)*, March 20-23, 2019, San Francisco, CA, USA.

[11] B. Abibullaev, Y. Orazayev, and A. Zollanvari, "**Novel Spatiospectral Features of ERPs Enhances Brain-Computer Interfaces**," in *Brain-Computer Interface (BCI), 2019 7th International Conference on*. IEEE Brain Society, 2019, GangWon, South Korea.

[12] B. Saduanov, D. Tokmurzina, T. Alizadeh, and B. Abibullaev, "**Brain-computer interface humanoid pre-trained for interaction with people**," in *2018 ACM/IEEE International Conference on Human-Robot Interaction*. ACM, March 5-8, 2018, pp. 229–230, Chicago, IL, USA.

[13] A. Oleinikov, B. Abibullaev, A. Shintemirov, and M. Folgheraiter, "**Feature extraction and real-time recognition of hand motion intentions from emgs via artificial neural networks**," in *Brain-Computer Interface (BCI), 2018 6th International Conference on*. IEEE Brain Society, 2018, pp. 1-5, GangWon, South Korea.

[14] G. Lee, S. H. Jin, S. T. Yang, J. An, and B. Abibullaev, "**Cross-correlation between HBO and HbR as an effective feature of motion artifact in the fNIRS signal**," 2018 6th International Conference on Brain-Computer Interface (BCI), pp. 1-3, 2018, IEEE Brain Society, GangWon, South Korea.

[15] B. Saduanov, T. Alizadeh, J. An, and B. Abibullaev, "**Trained by demonstration humanoid robot controlled via a BCI system for telepresence**," in *Brain-Computer Interface (BCI), 2018 6th International Conference on*. IEEE Brain Society, 2018, pp. 1–4, January, GangWon, South Korea.

[16] G. Lee, S. H. Jin, S. H. Lee, B. Abibullaev, and J. An, "**fNIRS motion artifact correction for overground walking using entropy-based unbalanced optode decision and wavelet regression neural network**," in *Multisensor Fusion and Integration for Intelligent Systems (M.F.I.), 2017 IEEE International Conference on*. IEEE, 2017, pp. 186–193, Daegu, South Korea.

[17] A. Zhumadilova, D. Tokmurzina, A. Kuderbekov, and B. Abibullaev. **Design and Evaluation of a P300 Visual Brain-Computer Interface Speller in Cyrillic Characters**. In *the 26th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN), IEEE*, 2017, Aug 28 - Sept 1, Lisbon, Portugal.

[18] B. Abibullaev. **Learning Suite of Kernel Feature Spaces Enhances SMR-Based EEG-BCI Classification.** In *the 5th International Winter Conference on Brain-Computer Interface, IEEE*, 2017, January 9-11, GangWon, South Korea.

[19] D. Nurseitov, A. Serekov, A. Shintemirov and B. Abibullaev. **Design and Evaluation of a P300-ERP based BCI System for Real-Time Control of a Mobile Robot.** In *the 5th International Winter Conference on Brain-Computer Interface, IEEE*, 2017, January 9-11, Gangwon , South Korea.

[20] B. Abibullaev and J An. **On Robust Classification of Hemodynamic Signals for BCIs via Multiple Kernel $\nu$-SVMs.** *Intelligent Robots and Systems (IROS), 2016 IEEE/RSJ International Conference*, 2016, Oct10-15, Daejeon, South Korea.

[21] J.G. Cruz-Garza, Z.R. Hernandez, M. Megjhani, B. Abibullaev, T. Tse, E. Caducoy, and JL Contreras-Vidal. **Neural Development of social cognition in the first two years of life - Early findings.** In *Society for Neuroscience* 2015 Oct 17-21; Chicago, USA.

[22] Z.R. Hernandez, A.J. Arenas-Castellanos, J.G. Cruz-Garza, M. Megjhani, B. Abibullaev, Sri. R.P. Maddi, T. Tse, C. Armstrong, W. Long, J.L. Contreras-Vidal **Decoding Intent From Non-invasive EEG in Freely Behaving Infants.** In *Ninth Biennial Meeting of the Cognitive Development Society* 2015 Oct 9-10; Ohio, U.S.A.

[23] A.J. Arenas-Castellanos, Z.R. Hernandez, J.G. Cruz-Garza, M. Megjhani, B. Abibullaev, Sri. R.P. Maddi, T. Tse, C. Armstrong, W. Long, J.L. Contreras-Vidal **A developmental Analysis of Behaviors Related to the Mirror Neuron System in 6-24 Months Infants** In *Ninth Biennial Meeting of the Cognitive Development Society* 2015 Oct 9-10; Ohio, U.S.A.

[24] Z.R. Hernandez, J.G. Cruz-Garza, T. Tse, E. Caducoy, B. Abibullaev, J.L. Contreras-Vidal. **Supervised Classification of Intended Behaviors Using Electroencephalography (EEG) from Freely-Behaving Infants: Early Findings.** In *12th Annual G.C.C. Conference on Theoretical and Computational Neuroscience*, Rice University, Houston, TX, United States, February 6-7, 2015.

[25] N. A. Bhagat, A Venkatakrishnan, B. Abibullaev, E. J. Artz, A.A. Blank, J. A. French, N. Yozbatiran, and R. G. Grossman, M. K. O'Malley, J. L. Contreras-Vidal, G. E. Francisco **BMI Control of a Therapeutic Exoskeleton to Facilitate Personalized Robotic Rehabilitation of the Upper Limb.** *In National Robotics Initiative (N.R.I.) P.I. Meeting*, Westin Arlington, Nov. 19-20 2014, United States.

[26] C.H.Park, B. Abibullaev, E.Y. Joo, S.C. Hong and S.B. Hong. **The evaluation of accuracy and clinical usefulness of 3D EEG source localization analysis.** In *the 19th Korean Epilepsy Congress*, Seoul, Republic of Korea, June 12-14 2014.

[27] S.H. Lee, J. An, G. Jang, S.H. Jin, B. Abibullaev, and J.I. Moon. **Neural activity during observation, imagery, and execution of eating: An fNIRS pilot study.** In *the 19th Annual Meeting of the Organization for Human Brain Mapping*, Seattle, WA, United States, June 16-20, 2013.

[28] J. An, S.H. Jin, S.H. Lee, G. Jang, B. Abibullaev, J. Ahn, H. Lee, and J.I. Moon. **Cortical activation pattern for grasping during observation, imagery, execution, FES, and observation-FES integrated BCI: An fNIRS pilot study.** In *the 35th Annual Int. Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Osaka, Japan, July 3-7 2013.

[29] J. An, S.H. Lee, S.H. Jin, B. Abibullaev, G. Jang, J. Ahn, H. Lee, and J.I. Moon. **The beginning of neurohaptics: Controlling cognitive interaction via brain haptic interface.** In *the 2013 IEEE International Winter Workshop on Brain-Computer Interface*, Gangwon Province, Korea, Feb. 18-20 2013.

[30] B. Abibullaev, J. An, J.I. Moon, S.H. Lee, and S.H. Jin. **A study on the BCI-Robot assisted stroke rehabilitation framework using brain hemodynamic signals.** In *the 9th Int Conference on Ubiquitous Robots and Ambient Intelligence*, IEEE proceedings, Daejon, Korea, November 26-29 2012.

[31] B. Abibullaev, J. An, S.H.Lee, S.H. Jin, J.I Moon, H.J. Lee. **Decoding of Hemodynamic Brain Responses for Brain-Computer Interfaces via Ensemble Support Vector Learning.** In *the Human-Computer Interaction Conference*, Gangwon-Do, Korea, January 11-13, 2012.

[32] B. Abibullaev, W.S. Kang, S.H. Lee, J. An, and H.D. Seo. **Near-Infrared Spectroscopy in the analysis of functional brain activity during cognitive tasks.** In *2010 IEEE Sensors Conference*, Hawaii, United States, November 1-4, 2010.

[33] B. Abibullaev, W.S. Kang, S.H. Lee, and J. An. **Recognition of brain hemodynamic mental response for brain-computer interfaces.** In *International Conference on Control Automation and Systems*, IEEE proceedings, Gyeonggi-do, Korea, October 27-30 2010.

[34] S.H. Lee, B. Abibullaev, W.S. Kang, and J. An. **analysis of attention deficit hyperactivity disorder in EEG using wavelet transform and self-organizing maps.** In *International Conference on Control Automation and Systems*, IEEE proceedings, Gyeonggi-do, Korea, October 27-30 2010.

[35] W.S. Kang, B. Abibullaev, S.H. Lee, and J. An. **Path planning algorithm using the values clustered by K-means.** In *the 15th Int. Symposium on Artificial Life and Robotics*, Japan, February 4-6, 2010.

[36] B. Abibullaev, H.D. Seo, W.S. Kang, and J. An. **A wavelet-based method for detecting and localizing epileptic neural spikes in EEG signals.** In *the 2nd Int. Conf. on Interaction Sciences: Information Technology, Culture, and Human. A.C.M.*, Seoul, Korea, Nov. 24– 26, 2009.

[37] B. Abibullaev and H.D. Seo. **Epileptic seizures detection using continuous time wavelet based neural networks.** In *the 6th International Conference on Information Technology: New Generations, IEEE Computer Society*, Las Vegas, Nevada, United States, April 27-29 2009.